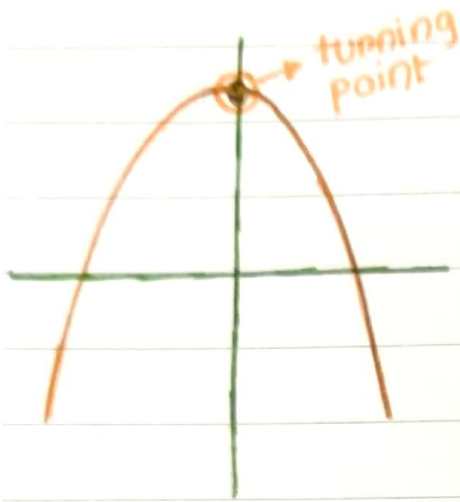


Rolle's Theorem



Rolle's theorem states that if a curve or a function intersects the x -axis twice, within a certain interval then there must be at least one turning point within that interval, provided that the function is continuous and differentiable.

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1. $f(x) = x^2 - 8x + 15$ $[3, 5]$

when $x = 3$,

$$(3)^2 - 8(3) + 15 = 0$$

when $x = 5$,

$$(5)^2 - 8(5) + 15 = 0$$

$$\therefore f'(x) = 2x - 8 = 0$$

$$\therefore x = 4$$

Some teachers ask students to write: "since the function is a polynomial, it is continuous and differentiable within the given interval" as starting statement

** Since $x = 4$ lies between

$(3, 5)$, Rolle's theorem is satisfied. [Ending statement]

2. $f(x) = \frac{1}{2}x - \sqrt{x}$ $[0, 4]$

$$\therefore f'(x) = \frac{1}{2} - \frac{1}{2\sqrt{x}} = 0$$

when $x = 0$,

$$\frac{1}{2} \times 0 - \sqrt{0} = 0$$

$$\Rightarrow \frac{\sqrt{x} - 1}{2\sqrt{x}} = 0$$

when $x = 4$,

$$\frac{1}{2} \times 4 - \sqrt{4} = 0$$

$$\Rightarrow \sqrt{x} = 1$$

$$\therefore x = 1$$

3. $f(x) = \cos x \quad [\pi/2, 3\pi/2]$

$$f(\pi/2) = \cos \pi/2 = 0$$

$$f(3\pi/2) = \cos 3\pi/2 = 0$$

$$f'(x) = -\sin x = 0$$

$$\therefore -\sin x = 0$$

$$x = 0$$

4. $f(x) = \ln(4 + 2x - x^2) \quad [-1, 3]$

$$f(-1) = \ln\{4 + 2(-1) - (-1)^2\} = 0$$

$$f(3) = \ln\{4 + 2(3) - (3)^2\} = 0$$

$$f'(x) = \frac{1}{(4 + 2x - x^2)} \cdot (2 - 2x)$$

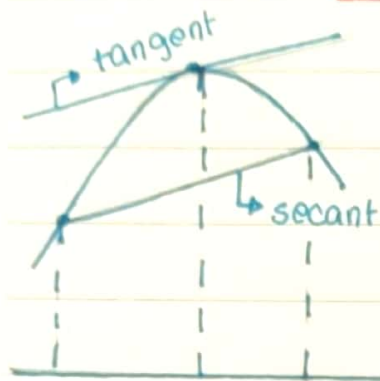
$$f'(x) = \frac{2(1-x)}{(4 + 2x - x^2)}$$

$$\text{Let } f'(x) = 0$$

$$2(1-x) = 0$$

$$\Rightarrow x = 1$$

Mean Value Theorem



Mean value theorem states that between any 2 points of a differential function (graph), there is at least one point on the graph where, if a tangent is drawn, it will be parallel to the secant line joining A the graph.

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5. $f(x) = x^2 - x$; $[-3, 5]$

$$f'(x) = 2x - 1$$

$$f(-3) = (-3)^2 - (-3) = 12$$

$$f(5) = (5)^2 - 5 = 20$$

Let $x = c$; $\therefore f'(c) = 2c - 1$

MVT Formula,

$$f'(c) = \frac{f(5) - f(-3)}{5 - (-3)}$$

$$\therefore 2c - 1 = \frac{20 - 12}{8}$$

$$\Rightarrow 16c - 8 = 20 - 12$$

$$\Rightarrow c = 1$$

Since $c = 1$ lies between $(-3, 5)$, MVT is verified.

MVT Formula:

$$f'(x) = \frac{f(b) - f(a)}{b - a}$$

6. $f(x) = x^3 + x - 4$; $[-1, 2]$

$$f'(x) = 3x^2 + 1$$

$$f(-1) = -6$$

$$f(2) = 6$$

Let $x = c$, $\therefore$ MVT formula,

$$3c^2 + 1 = \frac{6 - (-6)}{2 - (-1)}$$

$$\Rightarrow 3c^2 + 1 = \frac{12}{3}$$

$$\Rightarrow 3c^2 = 4 - 1$$

$$\Rightarrow c^2 = 1$$

Since $c = 1$ lies between $(-1, 2)$, MVT is verified

$$\underline{7.} \quad f(x) = \sqrt{25-x^2}; [-5, 3]$$

$$f'(x) = \frac{1}{2} \times (25-x^2)^{-1/2} \times -2x$$

$$= \frac{-x}{\sqrt{25-x^2}}$$

$$f(-5) = 0$$

$$f(3) = 4$$

$$\therefore \frac{-c}{\sqrt{25-c^2}} = \frac{4-0}{3-(-5)}$$

$$\therefore -2c = \sqrt{25-c^2}$$

$$\Rightarrow -4c^2 = 25-c^2$$

$$\Rightarrow -5c^2 = 25$$

$$\Rightarrow c = -5 - \sqrt{5}$$

Since $c = -\sqrt{5}$ lies between $(-5, 3)$, MVT is verified.

$$\underline{8.} \quad f(x) = x - \frac{1}{x}; [3, 4] \quad \Rightarrow \frac{c^2+1}{c^2} = \frac{13}{12}$$

$$f'(x) = 1 + \frac{1}{x^2}$$

$$\Rightarrow 12c^2 + 12 = 13c^2$$

$$\Rightarrow c = \sqrt{12}$$

$$f(3) = 8/3$$

$$\Rightarrow c = \pm 2\sqrt{3}$$

$$f(4) = 15/4$$

$$\therefore f'(c) = \frac{c^2+1}{c^2}$$

$$\therefore \frac{c^2+1}{c^2} = \frac{(15/4) - (8/3)}{4-3}$$